Chap 4
Applications of Differentiation

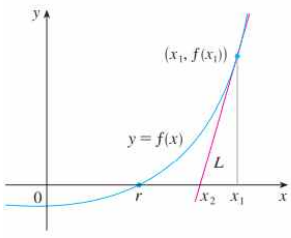
§6 Newton's Method

* We want to find the root r of the equation $f(x)=0$.

Let $y=f(x)$ and choose $x=x_1$.

$\Rightarrow$ The equation of the tangent of $y=f(x)$ at $(x_1, f(x_1))$ is

$$y-f(x_1)=f'(x_1)(x-x_1).$$



Let x_2 be the x-intercept of the tangent.

$$(x_2, 0): \; -f(x_1)=f'(x_1)(x_2-x_1)$$

$$\Rightarrow x_2=x_1-\frac{f(x_1)}{f'(x_1)}$$

Then x_2 is closer to r than x_1.

Using the tangent line at $(x_2, f(x_2))$, we have

$$x_3=x_2-\frac{f(x_2)}{f'(x_2)}$$

Repeating this process, we obtain

$$x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$$

where $\lim_{n\to\infty} x_n=r$.

<u>**Exam 1**</u> Starting with $x_1 = 2$, find the third approximation x_3 to the root of the equation $x^3 - 2x - 5 = 0$.

(sol)

<u>**Exam 2**</u> Use Newton's method to find $\sqrt[6]{2}$ correct to eight decimal places.

(sol)

Exam 3 Find, correct to six decimal places, the root of the equation $\cos x = x$.

(sol)